

MGTA 460: Business Analytics Project Management

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COURSE DESCRIPTION:

Analytics projects, though distinct from general projects, also share a large number of common challenges and pitfalls. Whether working for a startup designing and building the next-generation analytics tools to generate business insights, or for an established firm managing a small team executing an analytics project, competent project leadership requires understanding how to design, plan, and allocate work; how to coordinate and manage dispersed teams working on tightly interconnected pieces; how to manage senior management expectations; and how motivate and maintain focus of your project team. Good intentions cannot substitute for misdirected resource utilization and poor leadership, as is demonstrated by the fact that fully one-third of all projects fail and another third end up well below expectations.

While the general topic of project management is vast, this course has a narrower focus and will aim to provide relevant technical and management skills along three dimensions: (i) principles of *designing and planning* for analytics projects, (ii) *coordinating and integrating* work and managing its execution, and (iii) managing *risk and learning* to obtain desired outcomes through planned experimentation and exploration.

LEARNING OBJECTIVES:

- Task and process design (stage gate processes, checkins, milestones).
- Critical path method, Tools for scheduling and resource allocation.
- Risk management: valuing real options, experimentation & learning, flexibility
- Tools for coordinating team-work, designing information flows, modularity.
- Basics of agile and iterative development

COURSE MATERIALS:

A class-reader with a carefully selected set of readings has been designed for this course. In addition to the reader, I may hand out additional reading materials in class.

The assigned readings are given in the course schedule. Each student is expected to have completed the reading in preparation for each class before the class starts.

ATTENDANCE:

You are expected to attend every class; you are responsible for the material covered in class whether or not you attend. I realize that despite your best efforts you might be forced to miss a class due to unforeseen emergencies, and I would like you to inform me if you have (or will miss) a class.

TEAM WORK:

You shall work together with a team, and submit a joint assignment. The goal of this is not only to ensure that everyone in the class is on the same page, but more importantly, to ensure that you are using the collaboration tools that we will learn in this course. These tools will be useful for any team work for the rest of your program.

EVALUATION AND GRADING

1. 5 Pop-quizzes – 5x2=10pts
2. Dragonfly case (writeup and code) –30pts – (team assignment)
3. Final Exam – 40 pts
4. Two individual quizzes - 20 pts

Every class will have a 10 minute pop-quiz, to be completed in class. You will not receive any credit for unexcused absences. If you believe you will miss the class, it is your responsibility to inform me or the TA beforehand.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at: <http://www-senate.ucsd.edu/manual/appendices/app2.htm#AP14>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. No accommodations can be implemented retroactively.

Please visit the OSD website for further information or contact the Office for Students with Disabilities at (858) 534-4382 or fosorio@ucsd.edu.

SCHEDULE (TENTATIVE):

The following is a tentative schedule. In general, even if the specific date of coverage may change slightly, the order of coverage, and the associated material should remain as presented below.

Session	Topics	Notes	Readings
1	Basics of PM, planning and replanning, design and modularity.	Classroom exercise on design, iterations and modularity	1) The secrets to managing business Analytics Projects. HBS Product #SMR404-PDF-ENG 2) Why IT fumbles analytics. HBS Product #R1301H-PDF-ENG 3) What makes projects successful? HBS Product #2833BC-PDF-ENG
2	Scheduling, Crashing, Resource Allocation, and Tools.	Bring laptops with python installed on it. We will work through examples Quiz 1	4) A note on project management, HBS Product #W92C44-PDF-ENG
3	PERT and uncertainty, iterations in project, Matching the team to the task	Part A of Dragonfly assignment due before class 1-2 page design document due before class.	5) Dragonfly: Developing a proposal for an uninhabited aerial vehicle (UAV), INSEAD Case 600-003-1
4	Risk management, Experimentation, Real options	Quiz 2	6) The discipline of business experimentation, HBS Product #R1412D-PDF-ENG
5	Teamwork, collaboration tools (Basecamp), Principles of design		7) Jaume Ribera, "Agile Project Management," IESE Business School, Product No. PN-0505-E, 2018

Dragonfly Case & Class Project

You are expected to work in teams of 3-4 people to complete this team assignment.

PART A (Due prior to the 3rd session)

Assume for PART A that there is no uncertainty in the activity times given in the case (i.e., use the most likely estimate), and also assume for PART A that the interdependence among activities A4 and A9 does not exist (i.e., assume that while A9 depends on A4, A4 does not depend on A9). Each team will submit a single pdf file.

- 1) How could Bob Lake formally represent the evolution of the project, showing interdependence and parallelism among activities? What is the completion time of the project?
- 2) Should any activities be shortened to ensure timely completion of the proposal, and if so, which ones?

Design Document (Due prior to the 3rd session)

Along with submitting PART A of dragonfly case, you will also be required to submit a one-page design document for what you will deliver in Part B. This design document should clarify the goals, functional description, "interfaces," and milestones/checkpoints. See below for details on expectation from this design document. Each team will submit a single pdf file.

The design document, given the limited assignment scope, can be fairly short (even 1 or 2 pages). But at the minimum, it should address the following elements

1) *Goals*

Who will use it? Why will they use it? Provides a high-level description of its purpose.

2) *Description of functionality*

What does the tool/application do? Enumerate and clarify the use cases.

3) *Description of design and Interface.*

What functionality "chunks" will you have? How does the user interface with the tool? Do they type in values? Do they specify a csv file? Do they drag and drop to create the network?

4) *Checkpoints*

What are the intermediate deliverables and **when. Who** is responsible?

PART B (Due one week from last class)

For part B, you will need to create a tool that can be used to analyze uncertain activity durations in a project. To make things concrete, consider the Dragonfly case with uncertainty (and possible feedback loops). As a manager you may be interested to know the distribution of completion time (or costs), with or without crashing.

The deliverable for this part is a single python jupyter notebook that explicitly answers the following specific Dragonfly questions. In addition, the notebook should be flexible enough that I can upload a new csv data file with project information, and re-run the notebook to get new expected project duration and the histograms for the project duration for the newly uploaded project.

- 3) Now explicitly consider the uncertainty in activity durations, but assume that there is no interdependence between A4 and A9. Create a histogram to help understand the distribution of the project completion times. What is the expected completion time of the project?
- 4) How does the histogram change because of the interdependence between A4 and A9.
- 5) What is the expected cost of the proposal. Also, create a histogram to help understand the distribution of the cost.